

Introduction to Simulation of Plant Breeding Programs

Webinar Overview

This webinar provides an introduction to the simulation of plant breeding programs using **AlphaSimR**, an R package designed to simulate breeding programs at the genomic level. The course is aimed at introducing the main concepts and practical steps required to translate a breeding program into a simulation framework.

The focus will be on understanding how breeding decisions, population development, selection, crossing, and the passage of generations can be represented computationally. The simulations will be used as a framework to explore how different breeding strategies can affect the genetic trajectory and performance of a breeding program over time.

The course is intentionally **introductory and simplified**. The simulated breeding programs represent simplified versions of real breeding programs and do not attempt to reproduce all biological, operational, or statistical complexities encountered in practice. Likewise, simulations have inherent limitations related to the assumptions and parameters used to represent the genetic architecture, population, selection procedures, and breeding process.

Therefore, the simulations should not be interpreted as direct predictions of the performance of a real breeding program. Instead, they should be viewed as tools for understanding breeding program design and for exploring how alternative decisions can influence genetic change over time.

What We Will Cover

During the webinar, we will work through the main components required to build and evaluate a breeding program simulation.

We will begin by introducing the basic structure of **AlphaSimR**, including the creation of founder populations, simulation of genomes, definition of genetic traits, and specification of the parameters that control the simulation.

We will then construct and initialize breeding populations and discuss the concept of **burn-in**, which allows the simulated population to reach a more representative genetic state before evaluating future breeding strategies.

The main part of the course will focus on translating breeding pipelines into code. We will examine a **conventional breeding program for self-pollinated crops**, following the progression of populations through different breeding stages, selection, selfing, advancement, and variety development.

We will then modify the breeding pipeline to incorporate **genomic selection**, allowing us to compare alternative strategies and understand where genomic information can be incorporated into a breeding program.

We will also examine a **clonal breeding program**, including different approaches to selection and the consequences of changing the selection strategy within the breeding pipeline.

Finally, we will examine the results generated by the simulations and discuss how simulation can be used to compare breeding strategies and evaluate breeding program performance.

Throughout the course, the emphasis will be on understanding the connection between the **breeding program, the biological processes it represents, and the corresponding R code**.

Before the Webinar

To ensure that everyone can run the simulations during the webinar, please install **R** and the **AlphaSimR** package beforehand.

In R, run:

```
install.packages("AlphaSimR")
```

After installation, verify that AlphaSimR loads correctly:

```
library(AlphaSimR)
```

Please complete this installation **before** the webinar so that we can focus on the simulations and breeding program design during the session.

Additional Resources

For a broader introduction to plant breeding simulation with AlphaSimR, participants are encouraged to read **Bančić et al. (2024), “Plant breeding simulations with AlphaSimR.”** The paper provides a practical framework for designing stochastic breeding program simulations and includes examples involving self-pollinated, clonal, and hybrid crops, as well as breeding techniques such as genomic selection.

The **official AlphaSimR documentation** can be used as a reference throughout and after the webinar. It provides detailed information about the package functions, arguments, objects, and examples, and is particularly useful when adapting the simulations or developing more complex breeding programs.